



ODISHA UNIVERSITY OF TECHNOLOGY AND RESEARCH

Techno Campus, Mahalaxmi Vihar, Ghatikia, Bhubaneswar-751029.

Syllabus (Effective from 2023-24)

School/ Department: School of Computer Sciences

Course: M. Tech. (SSP), Programme: Computer Science and Engineering (CSE),

Duration: 2 years (Four Semesters)

2nd Semester

Sl. No.	Subject Type	Subject Code	Subject Name	Teaching Hours			Credit	Maximum Marks			
				L	T	P		IA	EA	PA	Total
1	PC 3	CS6102	High Performance Computing	3	0	0	3	40	60	-	100
2	PC 4	CS6104	Object Oriented Analysis and Design	3	0	0	3	40	60	-	100
3	PE 2 (Any One)	CS6202	Machine Learning Applications	3	0	0	3	40	60	-	100
		CS6204	Computer Graphics								
		CS6206	Mobile Computing								
4	PE 3 (Any One)	CS6208	Computer Vision	3	0	0	3	40	60	-	100
		CS6210	Cloud Computing								
		CS6212	Digital Forensics								
5	OE 1	Any One from the List of OE 1 (Appendix-I)		3	0	0	3	40	60	-	100
6	PR 1	CS6602	Project (Specialization Related)	0	0	4	2	-	-	100	100
7	LC 3	CS6502	Computing Lab - II	0	0	4	2	-	-	100	100
8	AC 2	Any One from the List of AC 2 (Appendix-I)		2	0	0	0	40	60	-	100
Total				17	0	8	19	240	360	200	800

3rd Semester

Sl. No.	Subject Type	Subject Code	Subject Name	Teaching Hours			Credit	Maximum Marks			
				L	T	P		IA	EA	PA	Total
1	PE 4* (Any One)	CS7201	Software Testing	3	0	0	3	40	60	-	100
		CS7203	Human Computer Interaction								
		CS7205	Real Time Systems								
2	PR 2	CS7601	Dissertation (Phase-I)	0	0	24	12	-	-	100	100
Total				3	0	24	15	40	60	100	200

* Virtual/Online Course either offered by OUTF or available in MOOCs platform (No physical class)

4th Semester

Sl. No.	Subject Type	Subject Code	Subject Name	Teaching Hours			Credit	Maximum Marks			
				L	T	P		IA	EA	PA	Total
1	PR 3	CS7602	Dissertation (Phase-II)	0	0	32	16	-	-	100	100
Total				0	0	32	16	-	-	100	100

Credits and Maximum Marks

Sl. No.	Semester	Credits	Maximum Marks
1	1 st	18	800
2	2 nd	19	800
3	3 rd	15	200
4	4 th	16	100
Total		68	1900



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3rd Semester

PE 4	CS7201	Software Testing	3	0	0	3
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Course Outcomes:

1. Analyze requirements to determine appropriate testing strategies.
2. Apply a wide variety of testing techniques in an effective and efficient manner.
3. Compute test coverage and yield according to a variety of criteria.
4. Evaluate the limitations of a given testing process and provide a succinct summary of those limitations.
5. Have an ability to identify the needs of software test automation, and define and develop a test tool to support test automation.
6. Have an ability understand and identify various software testing problems, and solve these problems by designing and selecting software test models, criteria, strategies, and methods
7. Have basic understanding and knowledge of contemporary issues in software testing, such as component-based software testing problems
8. Have an ability to use software testing methods and modern software testing tools for their testing projects.

Module-I:

INTRODUCTION: Testing as an Engineering Activity–Testing as a Process– Testing axioms–Basic definition– Software Testing Principles–The Tester’s Role in a Software Development Organization–Origins of Defects–Cost of defects–Defect Classes–The Defect Repository and Test Design–Defect Examples–Developer/Tester Support of Developing a Defect Repository –Defect Prevention strategies.

TESTCASEDESIGN: Test case Design Strategies– Using Black Box Approach to Test Case Design– Random Testing– Requirements based testing–Boundary Value Analysis–Equivalence Class Partitioning–State- based testing–Cause-effect graphing–Compatibility testing–user documentation testing–domain testing–Using WhiteBoxApproach to Test design– Test Adequacy Criteria–static testing vs. Structural testing–code functional testing–Coverage and Control Flow Graphs– Covering Code Logic– Paths–code complexity testing–Evaluating Test Adequacy Criteria.

Module-II:

LEVELSOFTESTING: The need for Levers of Testing–Unit Test –Unit Test Planning–Designing the Unit Tests–The Test Harness– Running the Unit tests and Recording results– Integration tests– Designing Integration Tests–Integration Test Planning–Scenario testing–Defect bash elimination

System Testing– Acceptance testing– Performance testing– Regression Testing– Inter nationalization testing– Ad-hoc testing–Alpha, Beta Tests–Testing OO systems–Usability and Accessibility testing–Configuration testing–Compatibility testing–Testing the documentation–Web site testing.

Module-III:

TESTMANAGEMENT: People and organizational issues in testing– Organization structures for testing teams– testing services– Test Planning – Test Plan Components– Test Plan Attachments– Locating Test Items– test management–test process–Reporting Test Results– The role of three groups in Test Planning and Policy Development–Introducing the test specialist–Skills needed by a test specialist–Building a Testing Group.

TEST AUTOMATION: Software test automation – skill needed for automation – scope of automation – design and architecture for automation–requirements for test tool–challenges in automation–Test metrics and measurements –project, progress and productivity metrics.

TEXTBOOKS:

1. Srinivasan Desikan and Gopalaswamy Ramesh, “Software Testing–Principles and Practices”, Pearson Education, 2006.
2. Ron Patton, “Software Testing”, Second Edition, Sams Publishing, Pearson Education, 2007.

REFERENCES:

1. Ilene Burnstein, “Practical Software Testing”, Springer International Edition, 2003.
2. Edward Kit, “Software Testing in the Real World – Improving the Process”, Pearson Education, 1995.
3. Boris Beizer, “Software Testing Techniques” – 2 nd Edition, Van Nostrand Reinhold, New York, 1990.
4. Aditya P. Mathur, “Foundations of Software Testing _ Fundamental Algorithms and Techniques”, Dorling Kindersley (India) Pvt. Ltd., Pearson Education, 2008.